A NOTE ON THE COMPARISON PRINCIPLE FOR DEGENERATE SUB-ELLIPTIC EQUATIONS: BLUEPRINT OF THE LEAN FORMALIZATION

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Abstract. This document collects the natural-language statements and detailed proofs that accompany the Lean 4 formalization of Theorem 1.3 and Corollary 1.4 of J. J. Manfredi and S. Mukherjee, *A note on the comparison principle for degenerate sub-elliptic equations*. The results are formalized for an arbitrary family of smooth vector fields on $\mathbb{R}^n$ satisfying Hörmander's rank condition; the paper's setting, a basis of the first layer of a Carnot group, is a special case. The mathematics is that of Section 3.1 of the paper; this document only reorganizes it into the units of the formalization. Each subsection corresponds to the Lean declaration of the same name in the directory `C2CompND/`; cross-references point to the results that a proof uses. Section 3 lists the formalized results of the paper. Within each section, items appear in dependency order.

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1. Definitions

1.1. `Preamble`.

Definition 1.1 (Ambient space, functions, and derivative conventions)**.** Throughout, $n, m \in \mathbb{N}$ and the ambient space is $\mathbb{R}^n$ equipped with the standard Euclidean inner product $\langle \cdot, \cdot \rangle$; vectors of $\mathbb{R}^m$ are written as functions $\{1, \dots, m\} \to \mathbb{R}$ and $\mathbb{R}^{m \times m}$ denotes the real $m \times m$ matrices.

Scalar functions and vector fields are taken to be defined on all of $\mathbb{R}^n$: a function of the paper's class $C^2(\Omega) \cap C(\overline{\Omega})$ is represented by a function $u : \mathbb{R}^n \to \mathbb{R}$ together with the hypotheses that u is of class C^2 on Ω and continuous on $\overline{\Omega}$. Since every hypothesis and every conclusion below is restricted to Ω or to $\overline{\Omega}$, and since Ω is open, the values of u outside $\overline{\Omega}$ never enter, and the derivatives of u at points of Ω are the intrinsic ones.

For $u : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathbb{R}^n$, $Du(x)$ denotes the Fréchet derivative of u at x, a linear functional on $\mathbb{R}^n$, identified with the Euclidean gradient through $Du(x)[z] = \langle \nabla u(x), z \rangle$. By convention $Du(x) := 0$ when u is not differentiable at x; the same convention is used for all the derivatives below. The second derivative is the bilinear form

$$D^2u(x)[z_1, z_2] := \Big(D\big(y \mapsto Du(y)\big)(x)[z_1] \Big)[z_2],$$

so that, in the matrix notation $D^2u = (\partial_{x_i} \partial_{x_j} u)_{i,j}$ of Section 2.1 of the paper,

$$D^2u(x)[z_1, z_2] = \langle D^2u(x)\, z_1, z_2 \rangle = \sum_{i,j=1}^{n} \partial_{x_i} \partial_{x_j} u(x)\, (z_1)_i\, (z_2)_j.$$

For a vector field $V : \mathbb{R}^n \to \mathbb{R}^n$, $DV(x)$ denotes its Jacobian, so that $DV(x)z$ is the derivative of V at x in the direction z.

We say that f is *of class C^k in a neighbourhood of x* if there is an open set containing x on which f is k times continuously differentiable, and *smooth* means of class C^k for every k. We write $g'(t)$ and $g''(t)$ for the first and second derivatives of a function $g : \mathbb{R} \to \mathbb{R}$, $\overline{S}$ for the closure of a set $S \subset \mathbb{R}^n$, ∂S for its topological frontier, and A^T for the transpose of a matrix A.

Definition 1.2 (Order on symmetric matrices and the Lie bracket)**.** For symmetric $A, B \in \mathbb{R}^{m \times m}$ we write $A \leq B$ when $B - A$ is positive semi-definite, i.e. $\langle (B - A)\xi, \xi \rangle \geq 0$ for every $\xi \in \mathbb{R}^m$; this is the order of Section 2.1 of the paper. A matrix is called *positive semi-definite* only if it is in addition symmetric, so that "A is positive semi-definite" always abbreviates "A is symmetric and $\langle A\xi, \xi \rangle \geq 0$ for every ξ".

For vector fields $V, W : \mathbb{R}^n \to \mathbb{R}^n$ the Lie bracket is the vector field

$$[V, W](x) := DW(x)\, V(x) - DV(x)\, W(x).$$

Proposition 1.3 (Weierstrass)**.** *Let X be a topological space, let $K \subset X$ be non-empty and compact and let $f : X \to \mathbb{R}$ be continuous on K. Then f attains a maximum over K: there is $x_0 \in K$ with $f(y) \leq f(x_0)$ for every $y \in K$. The same holds for the minimum.*

Proposition 1.4 (Closure of a bounded set, and the frontier decomposition)**.** *A bounded subset $\Omega \subset \mathbb{R}^n$ has compact closure $\overline{\Omega}$. For every subset $S \subset \mathbb{R}^n$ one has $\overline{S} = S \cup \partial S$.*

Proposition 1.5 (First-order necessary condition at a local extremum)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ and let x_0 be a local maximum point of f. Then $Df(x_0) = 0$. The same holds at a local minimum point.*

Proposition 1.6 (Symmetry of the second derivative)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 in a neighbourhood of x. Then the bilinear form $D^2f(x)$ is symmetric: $D^2f(x)[z_1, z_2] = D^2f(x)[z_2, z_1]$ for all $z_1, z_2 \in \mathbb{R}^n$.*

Proposition 1.7 (Second-derivative test, sufficient direction)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ and $t_0 \in \mathbb{R}$ be such that g is continuous at t_0, $g'(t_0) = 0$ and $g''(t_0) > 0$. Then t_0 is a local minimum point of g.*

Proposition 1.8 (Positive semi-definite square root)**.** *Let $A \in \mathbb{R}^{m\times m}$ be positive semi-definite. Then there is a positive semi-definite $P \in \mathbb{R}^{m\times m}$ with $P^2 = A$.*

Proposition 1.9 (Elementary properties of the trace)**.** *For $A, B, C \in \mathbb{R}^{m\times m}$ and $c \in \mathbb{R}$ one has $\mathrm{Tr}(A(B + C)) = \mathrm{Tr}(AB) + \mathrm{Tr}(AC)$, $\mathrm{Tr}(A(cB)) = c\,\mathrm{Tr}(AB)$, $\mathrm{Tr}(AB) = \sum_{i,k} A_{i,k}B_{k,i}$ and $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.*

1.2. `HGrad`.

Definition 1.10. Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ and let $X_i := \sigma^i(x) \cdot \nabla$ be the associated first-order operators, as in Section 2.1 of the paper. For $u : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathbb{R}^n$, the *horizontal (sub-elliptic) gradient* of u at x is the vector $\mathfrak{X}u(x) \in \mathbb{R}^m$ with components

$$\big(\mathfrak{X}u(x)\big)_i \;:=\; Du(x)\big[\sigma^i(x)\big] \;=\; \big\langle \nabla u(x), \sigma^i(x)\big\rangle \;=\; X_i u(x), \qquad i = 1, \dots, m,$$

where $Du(x)$ is the Fréchet derivative of u at x (equal to 0 when u is not differentiable at x, by the convention of the preamble).

This is the quantity $\mathfrak{X}u = (X_1u, \dots, X_mu)$ of Section 2.1 of the paper, equivalently $\mathfrak{X}u = \sigma(x)^T\nabla u$ in the notation of `eq:sigrad`, written as a function $\{1, \dots, m\} \to \mathbb{R}$.

1.3. `HHess`.

Definition 1.11. Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, with associated first-order operators $X_i = \sigma^i(x) \cdot \nabla$, and let $u : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathbb{R}^n$. The *horizontal second derivative matrix* of u at x is the matrix $\mathfrak{X}\mathfrak{X}u(x) \in \mathbb{R}^{m\times m}$ defined entrywise by

$$\big(\mathfrak{X}\mathfrak{X}u(x)\big)_{i,j} \;:=\; D\Big(y \mapsto Du(y)\big[\sigma^i(y)\big]\Big)(x)\big[\sigma^j(x)\big] \;=\; X_jX_iu(x), \qquad i, j = 1, \dots, m,$$

that is, the (i, j) entry is obtained by differentiating the function $y \mapsto X_iu(y) = \big(\mathfrak{X}u(y)\big)_i$ at x in the direction $\sigma^j(x)$. All derivatives follow the convention of the preamble (they are 0 where the relevant function fails to be differentiable).

Thus the (i, j) entry is X_jX_iu, matching the convention $\mathfrak{X}\mathfrak{X}u = (X_jX_iu)_{i,j}$ of Section 2.1 of the paper. This matrix is not symmetric in general.

1.4. `HHessSym`.

Definition 1.12. Let σ, u and x be as in `HHess`. The *symmetrized horizontal second derivative* of u at x is the matrix

$$\mathfrak{X}\mathfrak{X}^\star u(x) \;:=\; \tfrac{1}{2}\Big(\mathfrak{X}\mathfrak{X}u(x) + \big(\mathfrak{X}\mathfrak{X}u(x)\big)^T\Big) \;\in\; \mathbb{R}^{m\times m},$$

whose entries are $\big(\mathfrak{X}\mathfrak{X}^\star u(x)\big)_{i,j} = \frac{1}{2}\big(X_jX_iu(x) + X_iX_ju(x)\big)$. By construction $\mathfrak{X}\mathfrak{X}^\star u(x)$ is symmetric.

This is the quantity $\mathfrak{X}\mathfrak{X}^\star u = \frac{1}{2}\big(\mathfrak{X}\mathfrak{X}u + (\mathfrak{X}\mathfrak{X}u)^T\big) = \frac{1}{2}\big(X_iX_ju + X_jX_iu\big)_{i,j}$ of Section 2.1 of the paper.

1.5. `SubellipticOp`.

Definition 1.13. Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ and let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $a(\xi) = (a_{i,j}(\xi))_{i,j}$. For $u : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathbb{R}^n$ we set

$$\mathcal{L}u(x) \;:=\; -\,\mathrm{Tr}\Big(a\big(\mathfrak{X}u(x)\big)\,\mathfrak{X}\mathfrak{X}u(x)\Big) \;=\; -\sum_{i,j=1}^{m} a_{i,j}\big(\mathfrak{X}u(x)\big)\,X_iX_ju(x),$$

where $\mathfrak{X}u(x)$ is the horizontal gradient of `HGrad` and $\mathfrak{X}\mathfrak{X}u(x)$ the horizontal second derivative matrix of `HHess`; the second expression is the identity $\mathrm{Tr}(AB) = \sum_{i,k} A_{i,k}B_{k,i}$ together with $\big(\mathfrak{X}\mathfrak{X}u\big)_{j,i} = X_iX_ju$.

This is the operator of `eq:maineq` of the paper. Whenever $a(\xi)$ is symmetric the second expression coincides, after exchanging the summation indices, with the paper's $-\sum_{i,j} a_{i,j}(\mathfrak{X}u)X_jX_iu$.

1.6. `EllipticityFn`.

Definition 1.14. Let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $a(\xi) = (a_{i,j}(\xi))_{i,j}$. The associated *ellipticity function* $\mathcal{E} : \mathbb{R}^m \to \mathbb{R}$ is

$$\mathcal{E}(\xi) \;:=\; \big\langle a(\xi)\,\xi, \xi\big\rangle \;=\; \sum_{i,j=1}^{m} \xi_i\, a_{i,j}(\xi)\,\xi_j.$$

This is the quantity $\mathcal{E}(\xi) = \langle a(\xi)\xi, \xi\rangle$ appearing in the ellipticity condition `eq:ell` of the paper.

1.7. `IsClassicalSubsol`.

Definition 1.15. Let σ and a be as in `SubellipticOp`, let $U \subset \mathbb{R}^n$ and let $u : \mathbb{R}^n \to \mathbb{R}$. We say that u is a *classical sub-solution* of `eq:maineq` in U, and write $\mathcal{L}u \le 0$ in U, if

$$\mathcal{L}u(x) \le 0 \qquad \text{at every } x \in U \text{ with } \mathfrak{X}u(x) \neq 0.$$

Here $\mathfrak{X}u$ is the horizontal gradient of `HGrad` and $\mathcal{L}$ the operator of `SubellipticOp`. No condition is imposed at the horizontal critical points of u, where a need not be defined.

This is the sub-solution condition of Definition `def:classical` of the paper. The paper additionally takes U open and $u \in C^2(U)$; those are regularity requirements on U and u rather than part of the differential inequality, and the nodes that use this notion carry them as separate hypotheses.

1.8. `IsClassicalSupersol`.

Definition 1.16. Let σ and a be as in `SubellipticOp`, let $U \subset \mathbb{R}^n$ and let $u : \mathbb{R}^n \to \mathbb{R}$. We say that u is a *classical super-solution* of `eq:maineq` in U, and write $\mathcal{L}u \ge 0$ in U, if

$$\mathcal{L}u(x) \ge 0 \qquad \text{at every } x \in U \text{ with } \mathfrak{X}u(x) \neq 0.$$

Here $\mathfrak{X}u$ is the horizontal gradient of `HGrad` and $\mathcal{L}$ the operator of `SubellipticOp`. No condition is imposed at the horizontal critical points of u, where a need not be defined.

This is the super-solution condition of Definition `def:classical` of the paper, which defines sub- and super-solutions in one breath. It is not a restatement of the sub-solution condition of `IsClassicalSubsol` applied to $-u$: the matrix a is not assumed to be even (Remark `rem:hom`(a) of the paper says so explicitly), so in general $a(-\mathfrak{X}u) \neq a(\mathfrak{X}u)$ and $\mathcal{L}(-u) \neq -\mathcal{L}u$. As in `IsClassicalSubsol`, the paper's openness of U and C^2 regularity of u are carried as separate hypotheses by the nodes that use this notion.

1.9. `HorFamily`.

Definition 1.17. Let $n, m \in \mathbb{N}$ and let $\sigma = (\sigma^1, \dots, \sigma^m)$ be a family of vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$. We define HorFamily$(\sigma)$, a family of vector fields on $\mathbb{R}^n$, inductively by the two generation rules:

(i) $\sigma^i \in \mathrm{HorFamily}(\sigma)$ for every $i \in \{1, \dots, m\}$;
(ii) if $V, W \in \mathrm{HorFamily}(\sigma)$, then $[V, W] \in \mathrm{HorFamily}(\sigma)$, where $[V, W](x) = DW(x)\,V(x) - DV(x)\,W(x)$ is the Lie bracket.

Equivalently, HorFamily(σ) is the smallest family of vector fields on $\mathbb{R}^n$ that contains each σ^i and is closed under the Lie bracket.

This is the family of iterated commutators denoted $\mathrm{Lie}(X_1, \dots, X_m)$ in the paper's Hörmander rank condition `eq:horcond`, written for the first-order operators $X_i = \sigma^i(x) \cdot \nabla$ in terms of the coefficient fields σ^i of Section 2.1 of the paper.

1.10. `HormanderRank`.

Definition 1.18. Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be a family of vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$. We say that σ *satisfies the Hörmander rank condition*, written HormanderRank(σ), if for every $x \in \mathbb{R}^n$ the linear span of the set of evaluations at x of the vector fields of HorFamily(σ) (see `HorFamily`) is the whole of $\mathbb{R}^n$:

$$\mathrm{span}_{\mathbb{R}}\big\{V(x) \;:\; V \in \mathrm{HorFamily}(\sigma)\big\} = \mathbb{R}^n \qquad \text{for every } x \in \mathbb{R}^n.$$

Since $\mathbb{R}^n$ has dimension n, this says exactly that $\dim\big(\mathrm{Lie}(X_1, \dots, X_m)(x)\big) = n$ at every point $x \in \mathbb{R}^n$, which is the paper's condition `eq:horcond` for the first-order operators $X_i = \sigma^i(x) \cdot \nabla$.

2. Lemmas

2.1. `FDerivEqAtLocalMaxSub`.

Lemma 2.1. *Let $u, v : \mathbb{R}^n \to \mathbb{R}$ and let $x_0 \in \mathbb{R}^n$. Assume that u is differentiable at x_0, that v is differentiable at x_0, and that x_0 is a local maximum point of $u - v$. Then*

$$Du(x_0) = Dv(x_0)$$

as linear functionals on $\mathbb{R}^n$. This is the first clause of the sentence opening the proof of Lemma `lem:vweak` of the paper, namely that the interior maximality at x_0 implies $\nabla u(x_0) = \nabla v(x_0)$; the paper derives the horizontal identity $\mathfrak{X}u(x_0) = \mathfrak{X}v(x_0)$ from it through `eq:sigrad`.

Proof. Since u and v are differentiable at x_0, so is $u - v$, and the derivative of a difference is the difference of the derivatives:

$$D(u - v)(x_0) = Du(x_0) - Dv(x_0).$$

As x_0 is a local maximum point of $u - v$, the first-order necessary condition at a local extremum (preamble) applied to $f := u - v$ gives

$$D(u - v)(x_0) = 0.$$

Combining the two displays, $Du(x_0) - Dv(x_0) = 0$ in the vector space of linear functionals on $\mathbb{R}^n$, and therefore $Du(x_0) = Dv(x_0)$. □

2.2. `HGradEqAtLocalMax`.

Lemma 2.2. *Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ and let $x_0 \in \mathbb{R}^n$. Assume that u and v are differentiable at x_0 and that x_0 is a local maximum point of $u - v$. Then*

$$\mathfrak{X}u(x_0) = \mathfrak{X}v(x_0).$$

Proof. Since u and v are differentiable at x_0 and x_0 is a local maximum point of $u - v$, the hypotheses of `FDerivEqAtLocalMaxSub` are met, and it gives

$$Du(x_0) = Dv(x_0)$$

as linear functionals on $\mathbb{R}^n$.

Evaluating this equality of functionals at the vector $\sigma^i(x_0) \in \mathbb{R}^n$ gives, for every $i \in \{1, \dots, m\}$,

$$\big(\mathfrak{X}u(x_0)\big)_i = Du(x_0)\big[\sigma^i(x_0)\big] = Dv(x_0)\big[\sigma^i(x_0)\big] = \big(\mathfrak{X}v(x_0)\big)_i,$$

by the definition of the horizontal gradient (`HGrad`). Hence $\mathfrak{X}u(x_0) = \mathfrak{X}v(x_0)$. □

2.3. `DerivDerivRestrictLine`.

Lemma 2.3. *Let $n \in \mathbb{N}$, let $f : \mathbb{R}^n \to \mathbb{R}$, let $x_0 \in \mathbb{R}^n$ and let $w \in \mathbb{R}^n$. Assume that f is of class C^2 in a neighbourhood of x_0, and let $g : \mathbb{R} \to \mathbb{R}$, $g(t) := f(x_0 + tw)$, be the restriction of f to the line through x_0 in the direction w. Then*

$$g''(0) = D^2 f(x_0)[w, w].$$

Proof. Let $\gamma(t) := x_0 + tw$, so that $g = f \circ \gamma$. The map $\gamma : \mathbb{R} \to \mathbb{R}^n$ is affine, hence smooth, with $\gamma(0) = x_0$ and $\gamma'(t) = w$ for every t; being continuous with $\gamma(0) = x_0$, it satisfies $\gamma(t) \to x_0$ as $t \to 0$.

The first derivative of g near 0. By hypothesis there is an open set $V \ni x_0$ on which f is of class C^2; in particular f is differentiable at every point of V. Since $\gamma(t) \to x_0$ as $t \to 0$, there is a neighbourhood N of 0 with $\gamma(t) \in V$ for every $t \in N$. For such t the chain rule applies to $f \circ \gamma$ at t and gives

$$g'(t) = Df(\gamma(t))\big[\gamma'(t)\big] = Df(x_0 + tw)[w] \qquad \text{for every } t \in N. \tag{1}$$

Differentiating once more at 0. Since f is of class C^2 on V, the map $y \mapsto Df(y)$ is differentiable at x_0, with derivative the linear map $z \mapsto D\big(y \mapsto Df(y)\big)(x_0)[z]$. Composing it with the affine map γ (chain rule again, using $\gamma(0) = x_0$ and $\gamma'(0) = w$), the map $t \mapsto Df(x_0 + tw)$ is differentiable at $t = 0$ with derivative $D\big(y \mapsto Df(y)\big)(x_0)[w]$. Composing further with the

evaluation map $\Lambda \mapsto \Lambda[w]$, which is linear and hence smooth, the real-valued function $t \mapsto Df(x_0 + tw)[w]$ is differentiable at $t = 0$ with derivative

$$\Big(D\big(y \mapsto Df(y)\big)(x_0)[w]\Big)[w] = D^2 f(x_0)[w, w],$$

the last equality being the definition of $D^2 f(x_0)$ fixed in the preamble.

Conclusion. By (1) the functions g' and $t \mapsto Df(x_0 + tw)[w]$ agree on the neighbourhood N of 0, so they have the same derivative at 0. Hence g' is differentiable at 0 and $g''(0) = D^2 f(x_0)[w, w]$. □

2.4. `SecondDerivTestOneDim`.

Lemma 2.4. *Let $g : \mathbb{R} \to \mathbb{R}$ and $t_0 \in \mathbb{R}$. Assume that g is of class C^2 in a neighbourhood of t_0 and that t_0 is a local maximum point of g. Then*

$$g''(t_0) \leq 0.$$

Proof. Suppose, for contradiction, that $g''(t_0) > 0$.

Since g is of class C^2 near t_0, it is in particular differentiable at t_0, and t_0 is a local maximum point of g, so the first-order necessary condition (preamble) gives $g'(t_0) = 0$. Since g is continuous at t_0, $g'(t_0) = 0$ and $g''(t_0) > 0$, the sufficient direction of the second-derivative test (preamble) shows that t_0 is a *local minimum* point of g.

Thus t_0 is both a local maximum and a local minimum point of g: there are neighbourhoods N_1, N_2 of t_0 with $g \leq g(t_0)$ on N_1 and $g \geq g(t_0)$ on N_2. On the neighbourhood $N := N_1 \cap N_2$ of t_0 we therefore have $g \equiv g(t_0)$. Let $I \subset N$ be an open interval containing t_0. Then g is constant on I, hence $g' \equiv 0$ on I, i.e. g' coincides with the zero function on a neighbourhood of t_0. Differentiating, $g''(t_0) = (g')'(t_0) = 0$, which contradicts $g''(t_0) > 0$.

Hence $g''(t_0) \leq 0$. □

2.5. `SndFDerivNonposAtLocalMax`.

Lemma 2.5. *Let $n \in \mathbb{N}$, let $f : \mathbb{R}^n \to \mathbb{R}$ and let $x_0 \in \mathbb{R}^n$. Assume that f is of class C^2 in a neighbourhood of x_0 and that x_0 is a local maximum point of f. Then*

$$D^2 f(x_0)[w, w] \;\leq\; 0 \qquad \text{for every } w \in \mathbb{R}^n.$$

Proof. Fix $w \in \mathbb{R}^n$ and let $\gamma(t) := x_0 + tw$, $g := f \circ \gamma$, i.e. $g(t) = f(x_0 + tw)$ for $t \in \mathbb{R}$. The map $\gamma : \mathbb{R} \to \mathbb{R}^n$ is affine, hence smooth, with $\gamma(0) = x_0$ and $\gamma'(t) = w$ for every t.

Step 1: g is of class C^2 in a neighbourhood of 0. By hypothesis there is an open set $V \ni x_0$ on which f is of class C^2. Since γ is continuous and $\gamma(0) = x_0$, the set $\gamma^{-1}(V)$ is open and contains 0; on it $g = f \circ \gamma$ is the composition of a smooth map with a C^2 map, hence of class C^2.

Step 2: 0 is a local maximum point of g. By hypothesis there is a neighbourhood $N \ni x_0$ with $f(y) \leq f(x_0)$ for all $y \in N$. Then $\gamma^{-1}(N)$ is a neighbourhood of 0, and for $t \in \gamma^{-1}(N)$ we get $g(t) = f(\gamma(t)) \leq f(x_0) = g(0)$.

Step 3: $g''(0) = D^2 f(x_0)[w, w]$. The function f is of class C^2 in a neighbourhood of x_0, so `DerivDerivRestrictLine` applies to f, x_0 and w. Since g is precisely the restriction $t \mapsto f(x_0 + tw)$ of f to the line through x_0 in the direction w, its conclusion reads $g''(0) = D^2 f(x_0)[w, w]$.

Conclusion. By Steps 1 and 2 and `SecondDerivTestOneDim` applied to g at $t_0 = 0$, we have $g''(0) \leq 0$. By Step 3 this is exactly $D^2 f(x_0)[w, w] \leq 0$. □

2.6. `SndFDerivLeAtLocalMaxSub`.

Lemma 2.6. *Let $U \subset \mathbb{R}^n$ be open, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on U and let $x_0 \in U$ be a local maximum point of $u - v$. Then*

$$D^2 u(x_0)[w, w] \;\leq\; D^2 v(x_0)[w, w] \qquad \text{for every } w \in \mathbb{R}^n,$$

i.e. $D^2 u(x_0) \leq D^2 v(x_0)$. This is the second-order clause of the sentence opening the proof of Lemma `lem:vweak` of the paper, that the interior maximality at x_0 implies $\nabla u(x_0) = \nabla v(x_0)$ and $D^2 u(x_0) \leq D^2 v(x_0)$; the first-order clause of that same sentence is the node `FDerivEqAtLocalMaxSub`.

Proof. Put $f := u - v$. Since U is open, $x_0 \in U$ and u, v are of class C^2 on U, both u and v are of class C^2 in a neighbourhood of x_0, namely on U itself; hence so is f, and in particular u, v and f are differentiable at every point of U.

Step 1: the first derivatives subtract on U. For every $y \in U$ both u and v are differentiable at y, so $f = u - v$ is differentiable at y with

$$Df(y) = Du(y) - Dv(y) \qquad \text{for every } y \in U. \tag{1}$$

Since U is a neighbourhood of x_0, the maps $y \mapsto Df(y)$ and $y \mapsto Du(y) - Dv(y)$ therefore coincide on a neighbourhood of x_0.

Step 2: the second derivatives subtract at x_0. Because u and v are of class C^2 in a neighbourhood of x_0, the maps $y \mapsto Du(y)$ and $y \mapsto Dv(y)$ are differentiable at x_0, hence so is their difference. By Step 1 the map $y \mapsto Df(y)$ agrees with that difference on a neighbourhood of x_0, so it has the same derivative at x_0, and by linearity of the derivative

$$D^2 f(x_0)[z_1, z_2] = D^2 u(x_0)[z_1, z_2] - D^2 v(x_0)[z_1, z_2] \qquad \text{for all } z_1, z_2 \in \mathbb{R}^n. \tag{2}$$

Step 3: conclusion. The function f is of class C^2 in a neighbourhood of x_0 and has a local maximum at x_0, so `SndFDerivNonposAtLocalMax` applies to f and gives

$$D^2 f(x_0)[w, w] \le 0 \qquad \text{for every } w \in \mathbb{R}^n.$$

Fix $w \in \mathbb{R}^n$ and combine this with (2) taken at $z_1 = z_2 = w$:

$$D^2 u(x_0)[w, w] - D^2 v(x_0)[w, w] = D^2 f(x_0)[w, w] \le 0,$$

that is, $D^2 u(x_0)[w, w] \le D^2 v(x_0)[w, w]$. □

2.7. `HHessEntryFormula`.

Lemma 2.7. *Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, let $u : \mathbb{R}^n \to \mathbb{R}$, let $x \in \mathbb{R}^n$ and let $i, j \in \{1, \dots, m\}$. Assume that u is of class C^2 in a neighbourhood of x and that σ^i is of class C^1 in a neighbourhood of x. Then*

$$\big(\mathfrak{X}\mathfrak{X}u(x)\big)_{i,j} \;=\; D^2u(x)\big[\sigma^j(x), \sigma^i(x)\big] \;+\; Du(x)\big[D\sigma^i(x)\,\sigma^j(x)\big],$$

*that is, $X_j X_i u = \langle D^2 u\, \sigma^j, \sigma^i\rangle + \langle \nabla u, D\sigma^i\, \sigma^j\rangle$. This is the identity $X_j X_i u = \langle D^2 u\, \sigma^i, \sigma^j\rangle + \langle D\sigma^i\, \sigma^j, \nabla u\rangle$ recorded in Section 2.1 of the paper, which underlies **`eq:sigrad`** and the formula **`eq:Mxz`** for the first-order matrix $\mathcal{M}(x, \zeta)$.*

Proof. By the definition of the horizontal second derivative matrix (`HHess`),

$$\big(\mathfrak{X}\mathfrak{X}u(x)\big)_{i,j} = D\varphi(x)\big[\sigma^j(x)\big], \qquad \text{where} \qquad \varphi(y) := Du(y)\big[\sigma^i(y)\big].$$

It therefore suffices to compute $D\varphi(x)[z]$ for an arbitrary $z \in \mathbb{R}^n$ and then take $z = \sigma^j(x)$.

The function φ is the composition of the map $y \mapsto \big(Du(y), \sigma^i(y)\big)$ with the evaluation pairing

$$\Phi : \big(\mathbb{R}^n\big)^* \times \mathbb{R}^n \to \mathbb{R}, \qquad \Phi(\Lambda, \eta) := \Lambda[\eta],$$

which is bilinear and continuous. Its two components are differentiable at x:

- since u is of class C^2 in a neighbourhood of x, the map $y \mapsto Du(y)$ is differentiable at x, with derivative $z \mapsto D\big(y \mapsto Du(y)\big)(x)[z]$;
- since σ^i is of class C^1 in a neighbourhood of x, the map $y \mapsto \sigma^i(y)$ is differentiable at x, with derivative $z \mapsto D\sigma^i(x)\, z$.

By the chain rule applied to Φ composed with this pair (equivalently, by the product rule for a continuous bilinear pairing), φ is differentiable at x and for every $z \in \mathbb{R}^n$

$$D\varphi(x)[z] = \Big(D\big(y \mapsto Du(y)\big)(x)[z]\Big)\big[\sigma^i(x)\big] \;+\; Du(x)\big[D\sigma^i(x)\, z\big].$$

By the definition of the second derivative fixed in the preamble, the first summand is $D^2u(x)\big[z, \sigma^i(x)\big]$. Hence

$$D\varphi(x)[z] = D^2u(x)\big[z, \sigma^i(x)\big] + Du(x)\big[D\sigma^i(x)\, z\big].$$

Taking $z = \sigma^j(x)$ gives exactly the asserted identity. □

2.8. `PosSemidefSymmQuadForm`.

Lemma 2.8. *Let $n, m \in \mathbb{N}$, let $\mathcal{W}$ be a continuous bilinear form on $\mathbb{R}^n$ such that*

$$\mathcal{W}[z,z] \;\geq\; 0 \qquad \text{for every } z \in \mathbb{R}^n,$$

and let $s^1, \dots, s^m \in \mathbb{R}^n$ be arbitrary. Then the matrix $P \in \mathbb{R}^{m\times m}$ with entries

$$P_{i,j} := \tfrac{1}{2}\Big(\mathcal{W}\big[s^j, s^i\big] + \mathcal{W}\big[s^i, s^j\big]\Big)$$

is positive semi-definite, i.e. symmetric with $\langle P\xi, \xi\rangle \geq 0$ for every $\xi \in \mathbb{R}^m$.

Proof. Step 1: P is symmetric. Exchanging i and j in the defining formula exchanges the two summands and leaves their sum unchanged:

$$P_{j,i} = \tfrac{1}{2}\Big(\mathcal{W}\big[s^i, s^j\big] + \mathcal{W}\big[s^j, s^i\big]\Big) = P_{i,j}.$$

Step 2: expansion of the quadratic form. Fix $\xi \in \mathbb{R}^m$ and put $z := \sum_{i=1}^m \xi_i\, s^i \in \mathbb{R}^n$. By the definition of P and bilinearity of the summation,

$$\langle P\xi, \xi\rangle = \sum_{i,j=1}^m \xi_i\, P_{i,j}\, \xi_j = \tfrac{1}{2}\sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^j, s^i\big] + \tfrac{1}{2}\sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^i, s^j\big]. \tag{1}$$

Step 3: the two double sums agree. Renaming the pair of summation indices $i \leftrightarrow j$ in the first double sum of (1), and using $\xi_j\xi_i = \xi_i\xi_j$,

$$\sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^j, s^i\big] = \sum_{i,j=1}^m \xi_j\xi_i\, \mathcal{W}\big[s^i, s^j\big] = \sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^i, s^j\big]. \tag{2}$$

(Both double sums are finite, so the renaming is a bijective reindexing of the index set $\{1, \dots, m\}^2$.) Substituting (2) into (1), the two halves coincide and

$$\langle P\xi, \xi\rangle = \sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^j, s^i\big].$$

Step 4: conclusion. By bilinearity of $\mathcal{W}$, expanding z in both arguments,

$$\mathcal{W}[z,z] = \sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^i, s^j\big] = \sum_{i,j=1}^m \xi_i\xi_j\, \mathcal{W}\big[s^j, s^i\big],$$

the second equality being (2) read backwards. Hence $\langle P\xi, \xi\rangle = \mathcal{W}[z,z] \geq 0$ by hypothesis. Since $\xi \in \mathbb{R}^m$ was arbitrary, this together with the symmetry of Step 1 says that P is positive semi-definite. □

2.9. `HGradHHessComparison`.

Lemma 2.9. *Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ and let $x \in \mathbb{R}^n$. Assume that u and v are of class C^2 in a neighbourhood of x, that every σ^i is of class C^1 in a neighbourhood of x, and that*

$$Du(x) = Dv(x) \qquad \text{and} \qquad D^2u(x)[w,w] \leq D^2v(x)[w,w] \quad \text{for every } w \in \mathbb{R}^n.$$

Then

$$\mathfrak{X}u(x) = \mathfrak{X}v(x) \qquad \text{and} \qquad \mathfrak{X}\mathfrak{X}^\star v(x) - \mathfrak{X}\mathfrak{X}^\star u(x) \;\; \text{is positive semi-definite,}$$

the latter being $\mathfrak{X}\mathfrak{X}^\star u(x) \leq \mathfrak{X}\mathfrak{X}^\star v(x)$ in the order of Section 2.1 of the paper. This is the last sentence of Section 2.1 of the paper: if $u, w \in C^2$ satisfy $\nabla u(x) = \nabla w(x)$ and $D^2u(x) \leq D^2w(x)$ at a point x, then $\mathfrak{X}u(x) = \mathfrak{X}w(x)$ and $\mathfrak{X}\mathfrak{X}^\star u(x) \leq \mathfrak{X}\mathfrak{X}^\star w(x)$.

Proof. Step 1: the horizontal gradients. Evaluating the equality of linear functionals $Du(x) = Dv(x)$ at $\sigma^i(x)$ gives, for every i, $\big(\mathfrak{X}u(x)\big)_i = Du(x)[\sigma^i(x)] = Dv(x)[\sigma^i(x)] = \big(\mathfrak{X}v(x)\big)_i$ by the definition of `HGrad`. Hence $\mathfrak{X}u(x) = \mathfrak{X}v(x)$.

Step 2: the entries of the horizontal second derivative matrices. Fix i, j. The hypotheses of `HHessEntryFormula` hold for u (of class C^2 near x) and for v (of class C^2 near x), with σ^i of class C^1 near x. Therefore

$$\big(\mathfrak{X}\mathfrak{X}u(x)\big)_{i,j} = D^2u(x)\big[\sigma^j(x), \sigma^i(x)\big] + Du(x)\big[D\sigma^i(x)\,\sigma^j(x)\big],$$

$$\big(\mathfrak{X}\mathfrak{X}v(x)\big)_{i,j} = D^2v(x)\big[\sigma^j(x), \sigma^i(x)\big] + Dv(x)\big[D\sigma^i(x)\,\sigma^j(x)\big].$$

Since $Du(x) = Dv(x)$, the two first-order terms are equal, and subtracting gives

$$\big(\mathfrak{X}\mathfrak{X}v(x)\big)_{i,j} - \big(\mathfrak{X}\mathfrak{X}u(x)\big)_{i,j} = \mathcal{W}\big[\sigma^j(x), \sigma^i(x)\big], \qquad \mathcal{W} := D^2v(x) - D^2u(x), \tag{1}$$

where $\mathcal{W}$ is a bilinear form on $\mathbb{R}^n$.

Step 3: the difference of the symmetrized matrices. Write $P := \mathfrak{X}\mathfrak{X}^\star v(x) - \mathfrak{X}\mathfrak{X}^\star u(x)$ and $\Delta := \mathfrak{X}\mathfrak{X}v(x) - \mathfrak{X}\mathfrak{X}u(x)$. By the definition of the symmetrization (`HHessSym`) and linearity, $P = \frac{1}{2}\big(\Delta + \Delta^T\big)$, so that by (1)

$$P_{i,j} = \tfrac{1}{2}\big(\Delta_{i,j} + \Delta_{j,i}\big) = \tfrac{1}{2}\Big(\mathcal{W}\big[\sigma^j(x), \sigma^i(x)\big] + \mathcal{W}\big[\sigma^i(x), \sigma^j(x)\big]\Big) \tag{2}$$

for all $i, j \in \{1, \dots, m\}$.

Step 4: P is positive semi-definite. The form $\mathcal{W} = D^2v(x) - D^2u(x)$ is a continuous bilinear form on $\mathbb{R}^n$, being the difference of two second Fréchet derivatives, and for every $z \in \mathbb{R}^n$

$$\mathcal{W}[z, z] = D^2v(x)[z, z] - D^2u(x)[z, z] \;\geq\; 0$$

by the hypothesis $D^2u(x)[w, w] \leq D^2v(x)[w, w]$ applied with $w = z$. Hence `PosSemidefSymmQuadForm` applies to $\mathcal{W}$ and to the finite family $s^i := \sigma^i(x)$, $i = 1, \dots, m$, and asserts that the matrix with entries $\frac{1}{2}\big(\mathcal{W}[s^j, s^i] + \mathcal{W}[s^i, s^j]\big)$ is positive semi-definite. By (2) that matrix is P, so P is positive semi-definite, which with Step 1 completes the proof. □

2.10. `TraceMonotone`.

Lemma 2.10. *Let $m \in \mathbb{N}$ and let $A, X, Y \in \mathbb{R}^{m\times m}$. Assume that A is positive semi-definite and that $X - Y$ is positive semi-definite, i.e. $Y \leq X$ in the order on symmetric matrices of Section 2.1 of the paper. Then*

$$\mathrm{Tr}(AY) \;\leq\; \mathrm{Tr}(AX).$$

Proof. Set $Z := X - Y$, which is positive semi-definite by hypothesis. By linearity of the trace and distributivity of the matrix product (preamble),

$$\mathrm{Tr}(AX) - \mathrm{Tr}(AY) = \mathrm{Tr}\,\big(A(X - Y)\big) = \mathrm{Tr}(AZ),$$

so it suffices to prove $\mathrm{Tr}(AZ) \geq 0$.

Since A and Z are positive semi-definite, each has a positive semi-definite square root (preamble): there are symmetric $P, Q \in \mathbb{R}^{m\times m}$ with $P^2 = A$ and $Q^2 = Z$. Put $B := QP$, so that $B^T = P^TQ^T = PQ$. Using $\mathrm{Tr}(MN) = \mathrm{Tr}(NM)$ (preamble) with $M = PPQ$ and $N = Q$ we get

$$\mathrm{Tr}(AZ) = \mathrm{Tr}(PPQQ) = \mathrm{Tr}(QPPQ) = \mathrm{Tr}\,\big((QP)(PQ)\big) = \mathrm{Tr}\,\big(BB^T\big).$$

Finally, by the entrywise formula for the trace of a product (preamble),

$$\mathrm{Tr}\,\big(BB^T\big) = \sum_{i,k=1}^{m} B_{i,k}\,\big(B^T\big)_{k,i} = \sum_{i,k=1}^{m} B_{i,k}^2 \;\geq\; 0.$$

Hence $\mathrm{Tr}(AX) - \mathrm{Tr}(AY) = \mathrm{Tr}(AZ) \geq 0$, which is the claim. □

2.11. `TraceMulSymmetrizeEq`.

Lemma 2.11. *Let $m \in \mathbb{N}$, let $A \in \mathbb{R}^{m\times m}$ be symmetric, i.e. $A^T = A$, and let $B \in \mathbb{R}^{m\times m}$ be arbitrary. Then*

$$\operatorname{Tr}\Big(A \cdot \tfrac{1}{2}\big(B + B^T\big)\Big) = \operatorname{Tr}(AB).$$

Equivalently, replacing the second factor by its symmetrization does not change its trace against a symmetric matrix. This is the matrix fact underlying the paper's remark, just after `eq:maineq`*, that because $a(\xi)$ is symmetric the equation $\mathcal{L}u = -\operatorname{Tr}\big(a(\mathfrak{X}u)\mathfrak{X}\mathfrak{X}u\big) = 0$ can also be written with the symmetrized sub-elliptic second derivative $\mathfrak{X}\mathfrak{X}^\star u$ in place of $\mathfrak{X}\mathfrak{X}u$.*

Proof. First, $\operatorname{Tr}\big(AB^T\big) = \operatorname{Tr}(AB)$. Indeed, by the entrywise formula for the trace of a product (preamble) and the symmetry of A,

$$\operatorname{Tr}\big(AB^T\big) = \sum_{i,k=1}^{m} A_{i,k}\big(B^T\big)_{k,i} = \sum_{i,k=1}^{m} A_{i,k}B_{i,k} = \sum_{i,k=1}^{m} A_{k,i}B_{i,k} = \operatorname{Tr}(AB),$$

where the first and last equalities are that formula applied to AB^T and to AB, the second uses $\big(B^T\big)_{k,i} = B_{i,k}$, and the third renames the summation indices $i \leftrightarrow k$ and uses $A_{i,k} = A_{k,i}$, which is the hypothesis $A^T = A$.

Hence, using the linearity of the trace in the sense of the identities $\operatorname{Tr}\big(A(B+C)\big) = \operatorname{Tr}(AB) + \operatorname{Tr}(AC)$ and $\operatorname{Tr}\big(A(cB)\big) = c\operatorname{Tr}(AB)$ (preamble), with $C := B^T$ and $c := \frac{1}{2}$,

$$\operatorname{Tr}\Big(A\cdot\tfrac{1}{2}\big(B+B^T\big)\Big) = \tfrac{1}{2}\operatorname{Tr}\big(A(B+B^T)\big) = \tfrac{1}{2}\operatorname{Tr}(AB) + \tfrac{1}{2}\operatorname{Tr}\big(AB^T\big) = \tfrac{1}{2}\operatorname{Tr}(AB) + \tfrac{1}{2}\operatorname{Tr}(AB) = \operatorname{Tr}(AB),$$

the third equality by the first paragraph. □

2.12. `LocalMaxOpIneq`.

Lemma 2.12. *Let $U \subset \mathbb{R}^n$ be open, let $\sigma = (\sigma^1, \dots, \sigma^m)$ be smooth vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ and let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ be such that $a(\xi)$ is positive semi-definite (hence symmetric) for every $\xi \in \mathbb{R}^m \setminus \{0\}$. Let $u, v : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on U and let $x_0 \in U$ be a local maximum point of $u - v$ with $\mathfrak{X}u(x_0) \neq 0$. Then*

$$\mathfrak{X}v(x_0) = \mathfrak{X}u(x_0), \qquad \mathfrak{X}\mathfrak{X}^\star v(x_0) - \mathfrak{X}\mathfrak{X}^\star u(x_0) \ \text{ is positive semi-definite,} \qquad \mathcal{L}v(x_0) \le \mathcal{L}u(x_0),$$

the middle assertion being $\mathfrak{X}\mathfrak{X}^\star u(x_0) \le \mathfrak{X}\mathfrak{X}^\star v(x_0)$ in the order of Section 2.1 of the paper. This is the first assertion of Lemma `lem:vweak` *of the paper.*

Proof. Since U is open and $x_0 \in U$, and u, v are of class C^2 on U, both u and v are of class C^2 in a neighbourhood of x_0 (namely on U itself); in particular both are differentiable at x_0. Each σ^i is smooth, hence of class C^1 in a neighbourhood of x_0.

Step 1: $\mathfrak{X}v(x_0) = \mathfrak{X}u(x_0)$. Apply `HGradEqAtLocalMax` to σ, u, v and x_0: its hypotheses are that u and v are differentiable at x_0 and that x_0 is a local maximum point of $u - v$, all of which hold. It gives $\mathfrak{X}u(x_0) = \mathfrak{X}v(x_0)$. Write $\xi_0 := \mathfrak{X}u(x_0) = \mathfrak{X}v(x_0)$; by hypothesis $\xi_0 \neq 0$.

Step 2: $Du(x_0) = Dv(x_0)$. Apply `FDerivEqAtLocalMaxSub` to u, v and x_0: its hypotheses are that u and v are differentiable at x_0 and that x_0 is a local maximum point of $u - v$, all of which hold. It gives $Du(x_0) = Dv(x_0)$.

Step 3: $D^2u(x_0)[w,w] \le D^2v(x_0)[w,w]$ for every $w \in \mathbb{R}^n$. Apply `SndFDerivLeAtLocalMaxSub` to U, u, v and x_0: its hypotheses are that U is open, that $x_0 \in U$, that u and v are of class C^2 on U and that x_0 is a local maximum point of $u - v$, all of which hold by hypothesis. It gives exactly the claim.

Step 4: the symmetrized horizontal second derivatives. Steps 2 and 3 are exactly the two hypotheses of `HGradHHessComparison`, whose remaining hypotheses (u, v of class C^2 near x_0, each σ^i of class C^1 near x_0) also hold. It yields that

$$P := \mathfrak{X}\mathfrak{X}^\star v(x_0) - \mathfrak{X}\mathfrak{X}^\star u(x_0)$$

is positive semi-definite, which is the second assertion.

Step 5: the trace comparison. Put $A := a(\xi_0)$. Since $\xi_0 \neq 0$, A is positive semi-definite by hypothesis; in particular A is symmetric. Apply `TraceMonotone` with this A, with $X :=$

$\mathfrak{X}\mathfrak{X}^\star v(x_0)$ and $Y := \mathfrak{X}\mathfrak{X}^\star u(x_0)$: its hypotheses are that A is positive semi-definite (just checked) and that $X - Y = P$ is positive semi-definite (Step 4). It gives

$$\operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}^\star u(x_0)\Big) \;\le\; \operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}^\star v(x_0)\Big). \tag{2}$$

Step 6: symmetrization does not change the trace against a symmetric matrix. The matrix $A = a(\xi_0)$ is symmetric (Step 5), so `TraceMulSymmetrizeEq` applies to A and to every $B \in \mathbb{R}^{m\times m}$ and gives $\operatorname{Tr}\big(A \cdot \frac{1}{2}(B + B^T)\big) = \operatorname{Tr}(AB)$. Taking $B = \mathfrak{X}\mathfrak{X}u(x_0)$ and then $B = \mathfrak{X}\mathfrak{X}v(x_0)$, and recalling that by the definition of the symmetrization (`HHessSym`) one has $\mathfrak{X}\mathfrak{X}^\star w(x_0) = \frac{1}{2}\big(\mathfrak{X}\mathfrak{X}w(x_0) + \mathfrak{X}\mathfrak{X}w(x_0)^T\big)$,

$$\operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}^\star u(x_0)\Big) = \operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}u(x_0)\Big), \qquad \operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}^\star v(x_0)\Big) = \operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}v(x_0)\Big). \tag{3}$$

This is the paper's remark, just after `eq:maineq`, that the equation can equivalently be written with the symmetrized horizontal second derivative.

Step 7: conclusion. By the definition of the operator (`SubellipticOp`) and $\mathfrak{X}u(x_0) = \mathfrak{X}v(x_0) = \xi_0$,

$$\mathcal{L}u(x_0) = -\operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}u(x_0)\Big), \qquad \mathcal{L}v(x_0) = -\operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}v(x_0)\Big).$$

Combining this with (3) and then (2),

$$-\mathcal{L}u(x_0) = \operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}^\star u(x_0)\Big) \;\le\; \operatorname{Tr}\Big(A\,\mathfrak{X}\mathfrak{X}^\star v(x_0)\Big) = -\mathcal{L}v(x_0),$$

that is, $\mathcal{L}v(x_0) \le \mathcal{L}u(x_0)$. □

2.13. `SndFDerivCompScalar`.

Lemma 2.13. *Let $n \in \mathbb{N}$, let $h \in C^2(\mathbb{R})$, let $w : \mathbb{R}^n \to \mathbb{R}$ and let $x \in \mathbb{R}^n$. Assume that w is of class C^2 in a neighbourhood of x. Then, for all $z_1, z_2 \in \mathbb{R}^n$,*

$$D^2(h\circ w)(x)[z_1,z_2] = h''\big(w(x)\big)\,Dw(x)[z_1]\,Dw(x)[z_2] + h'\big(w(x)\big)\,D^2w(x)[z_1,z_2].$$

Proof. Write $t := h'(w(x))$ and $s := h''(w(x))$, and let V be an open neighbourhood of x on which w is of class C^2.

Step 1: the first derivative of $h\circ w$ on V. Since $h \in C^2(\mathbb{R})$, h is differentiable on $\mathbb{R}$; and w is differentiable at every $y \in V$. Hence the chain rule applies to $h\circ w$ at every $y \in V$ and gives

$$D(h\circ w)(y) = h'\big(w(y)\big)\,Dw(y) \qquad \text{for every } y \in V. \tag{1}$$

In particular the map $y \mapsto D(h\circ w)(y)$ agrees on the neighbourhood V of x with the product of the scalar function $y \mapsto h'(w(y))$ and the $(\mathbb{R}^n)^*$-valued function $y \mapsto Dw(y)$.

Step 2: the two factors are differentiable at x. Since $h \in C^2(\mathbb{R})$, the function h' is differentiable on $\mathbb{R}$, and w is differentiable at x; so by the chain rule $y \mapsto h'(w(y))$ is differentiable at x with

$$D\big(y \mapsto h'(w(y))\big)(x)[z_1] = h''\big(w(x)\big)\,Dw(x)[z_1] = s\,Dw(x)[z_1] \qquad \text{for every } z_1 \in \mathbb{R}^n.$$

Since w is of class C^2 on V, the map $y \mapsto Dw(y)$ is differentiable at x, with derivative $z_1 \mapsto D\big(y \mapsto Dw(y)\big)(x)[z_1]$.

Step 3: the product rule. By Step 2 and the product rule for a scalar function times a vector-valued map, the product of the two factors is differentiable at x, and by Step 1 it agrees with $y \mapsto D(h\circ w)(y)$ on a neighbourhood of x; hence that map is differentiable at x with the same derivative, which, using (1) at $y = x$ to evaluate the scalar factor, is

$$D\big(y \mapsto D(h\circ w)(y)\big)(x)[z_1] = s\,\big(Dw(x)[z_1]\big)\,Dw(x) + t\,D\big(y \mapsto Dw(y)\big)(x)[z_1]$$

for every $z_1 \in \mathbb{R}^n$. Evaluating this identity of linear functionals at $z_2 \in \mathbb{R}^n$ and using the definition of the second derivative from the preamble, i.e. $D^2g(x)[z_1,z_2] = \big(D(y \mapsto Dg(y))(x)[z_1]\big)[z_2]$,

$$D^2(h\circ w)(x)[z_1,z_2] = s\,Dw(x)[z_1]\,Dw(x)[z_2] + t\,D^2w(x)[z_1,z_2],$$

which is the claim. □

2.14. `HGradHHessComp`.

Lemma 2.14. *Let $\sigma = (\sigma^1, \dots, \sigma^m)$ be vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, let $h \in C^2(\mathbb{R})$, let $w : \mathbb{R}^n \to \mathbb{R}$ and let $x \in \mathbb{R}^n$. Assume that w is of class C^2 in a neighbourhood of x and that every σ^i is of class C^1 in a neighbourhood of x. Then*

$$\mathfrak{X}(h \circ w)(x) = h'\big(w(x)\big)\, \mathfrak{X}w(x)$$

and

$$\mathfrak{X}\mathfrak{X}(h \circ w)(x) = h'\big(w(x)\big)\, \mathfrak{X}\mathfrak{X}w(x) + h''\big(w(x)\big)\, \mathfrak{X}w(x) \otimes \mathfrak{X}w(x),$$

where $\xi \otimes \eta \in \mathbb{R}^{m\times m}$ denotes the matrix with entries $(\xi \otimes \eta)_{i,j} = \xi_i\, \eta_j$. These are the two identities $\mathfrak{X}(h \circ w) = h'(w)\mathfrak{X}w$ and $\mathfrak{X}\mathfrak{X}(h \circ w) = h'(w)\mathfrak{X}\mathfrak{X}w + h''(w)\, \mathfrak{X}w \otimes \mathfrak{X}w$ with which the proof of Lemma `lem:tech` of the paper begins.

Proof. Write $t := h'(w(x))$, $s := h''(w(x))$ and $\xi := \mathfrak{X}w(x)$.

Step 1: the first identity. Let V be an open neighbourhood of x on which w is of class C^2. Since h is differentiable on $\mathbb{R}$ and w is differentiable on V, the chain rule gives that $h \circ w$ is differentiable on V with

$$D(h \circ w)(y) = h'\big(w(y)\big)\, Dw(y) \qquad \text{for every } y \in V. \tag{1}$$

Evaluating (1) at $y = x$ and at the vector $\sigma^i(x)$, and using the definition of `HGrad`,

$$\big(\mathfrak{X}(h \circ w)(x)\big)_i = D(h \circ w)(x)\big[\sigma^i(x)\big] = t\, Dw(x)\big[\sigma^i(x)\big] = t\, \xi_i,$$

for every i, which is the first identity.

Step 2: the second derivative of $h \circ w$. Since $h \in C^2(\mathbb{R})$ and w is of class C^2 on V, the composition $h \circ w$ is of class C^2 on V, hence in a neighbourhood of x. Moreover $h \in C^2(\mathbb{R})$ and w is of class C^2 in a neighbourhood of x, which are the hypotheses of `SndFDerivCompScalar`; applied to h, w and x it gives

$$D^2(h \circ w)(x)[z_1, z_2] = s\, Dw(x)[z_1]\, Dw(x)[z_2] + t\, D^2w(x)[z_1, z_2] \tag{2}$$

for all $z_1, z_2 \in \mathbb{R}^n$.

Step 3: the second identity. Fix i, j. By Step 2 the function $h \circ w$ is of class C^2 in a neighbourhood of x, and σ^i is of class C^1 in a neighbourhood of x; so `HHessEntryFormula` applies to $h \circ w$ and gives

$$\big(\mathfrak{X}\mathfrak{X}(h \circ w)(x)\big)_{i,j} = D^2(h \circ w)(x)\big[\sigma^j(x), \sigma^i(x)\big] + D(h \circ w)(x)\big[D\sigma^i(x)\, \sigma^j(x)\big].$$

By (2) with $z_1 = \sigma^j(x)$, $z_2 = \sigma^i(x)$, and by the definition of `HGrad`, the first summand equals

$$s\, Dw(x)\big[\sigma^j(x)\big]\, Dw(x)\big[\sigma^i(x)\big] + t\, D^2w(x)\big[\sigma^j(x), \sigma^i(x)\big] = s\, \xi_j\, \xi_i + t\, D^2w(x)\big[\sigma^j(x), \sigma^i(x)\big],$$

while by (1) the second summand equals $t\, Dw(x)\big[D\sigma^i(x)\, \sigma^j(x)\big]$. Adding and applying `HHessEntryFormula` to w itself (which is of class C^2 near x),

$$\big(\mathfrak{X}\mathfrak{X}(h\circ w)(x)\big)_{i,j} = t\Big(D^2w(x)\big[\sigma^j(x), \sigma^i(x)\big] + Dw(x)\big[D\sigma^i(x)\, \sigma^j(x)\big]\Big) + s\, \xi_i\, \xi_j = t\, \big(\mathfrak{X}\mathfrak{X}w(x)\big)_{i,j} + s\, \xi_i\, \xi_j.$$

Since $(\xi \otimes \xi)_{i,j} = \xi_i \xi_j$, this is precisely the second identity entrywise, and hence as an identity of matrices. □

2.15. `PerturbationIdentity`.

Lemma 2.15. *Let $U \subset \mathbb{R}^n$ be open, let $\sigma = (\sigma^1, \dots, \sigma^m)$ be smooth vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, let $\beta \in \mathbb{R}$ and let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ satisfy the homogeneity condition*

$$a(t\xi) = t^\beta\, a(\xi) \qquad \text{for every } t > 0 \text{ and every } \xi \in \mathbb{R}^m \setminus \{0\}.$$

Let $h \in C^2(\mathbb{R})$, let $w : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on U and let $x \in U$ be such that $\mathfrak{X}w(x) \neq 0$ and $h'(w(x)) > 0$. Then

$$\mathcal{L}(h \circ w)(x) \;=\; -h'\big(w(x)\big)^\beta \Big[h''\big(w(x)\big)\, \mathcal{E}\big(\mathfrak{X}w(x)\big) - h'\big(w(x)\big)\, \mathcal{L}w(x)\Big],$$

where $\mathcal{E}$ is the ellipticity function of `EllipticityFn` and t^β denotes the real power of the positive real number $t = h'(w(x))$. This is Lemma `lem:tech` (identity `eq:estLh`) of the paper.

Proof. Put $t := h'(w(x)) > 0$, $s := h''(w(x))$ and $\xi := \mathfrak{X}w(x) \neq 0$.

Since U is open, $x \in U$ and w is of class C^2 on U, the function w is of class C^2 in a neighbourhood of x; and each σ^i is smooth, hence of class C^1 in a neighbourhood of x.

Step 1: the two chain rules. By `HGradHHessComp` (whose hypotheses are exactly $h \in C^2(\mathbb{R})$, w of class C^2 near x, and each σ^i of class C^1 near x),

$$\mathfrak{X}(h \circ w)(x) = t\,\xi, \qquad \mathfrak{X}\mathfrak{X}(h \circ w)(x) = t\,\mathfrak{X}\mathfrak{X}w(x) + s\,\xi \otimes \xi. \tag{1}$$

Step 2: homogeneity of a at the perturbed gradient. Since $t > 0$ and $\xi \neq 0$, the homogeneity hypothesis applies and gives

$$a\big(\mathfrak{X}(h \circ w)(x)\big) = a(t\,\xi) = t^{\beta}\,a(\xi). \tag{2}$$

Step 3: expanding the operator. By the definition of $\mathcal{L}$ (`SubellipticOp`) applied to $h \circ w$, then by (1), (2) and the linearity of the trace in its second factor (preamble),

(3)

$$\mathcal{L}(h\circ w)(x) = -\operatorname{Tr}\Big(a\big(\mathfrak{X}(h\circ w)(x)\big)\,\mathfrak{X}\mathfrak{X}(h\circ w)(x)\Big) = -t^{\beta}\Big[t\operatorname{Tr}\big(a(\xi)\,\mathfrak{X}\mathfrak{X}w(x)\big) + s\operatorname{Tr}\big(a(\xi)\,\xi\otimes\xi\big)\Big].$$

Step 4: the two traces. First, by the entrywise formula for the trace of a product (preamble) and $(\xi \otimes \xi)_{k,i} = \xi_k \xi_i$,

$$\operatorname{Tr}\big(a(\xi)\,\xi \otimes \xi\big) = \sum_{i,k=1}^{m} a_{i,k}(\xi)\,(\xi \otimes \xi)_{k,i} = \sum_{i,k=1}^{m} \xi_i\,a_{i,k}(\xi)\,\xi_k = \big\langle a(\xi)\xi, \xi\big\rangle = \mathcal{E}(\xi),$$

by the definition of `EllipticityFn`. Second, since $\mathfrak{X}w(x) = \xi$, the definition of $\mathcal{L}$ (`SubellipticOp`) applied to w at x reads $\mathcal{L}w(x) = -\operatorname{Tr}\big(a(\xi)\,\mathfrak{X}\mathfrak{X}w(x)\big)$, i.e.

$$\operatorname{Tr}\big(a(\xi)\,\mathfrak{X}\mathfrak{X}w(x)\big) = -\mathcal{L}w(x).$$

Step 5: conclusion. Substituting the two identities of Step 4 into (3),

$$\mathcal{L}(h \circ w)(x) = -t^{\beta}\big[-t\,\mathcal{L}w(x) + s\,\mathcal{E}(\xi)\big] = -t^{\beta}\big[s\,\mathcal{E}(\xi) - t\,\mathcal{L}w(x)\big],$$

which, with $t = h'(w(x))$, $s = h''(w(x))$ and $\xi = \mathfrak{X}w(x)$, is the asserted identity. □

2.16. `MaxSetStability`.

Lemma 2.16. *Let X be a topological space, let $K \subset X$ be non-empty and compact, let $f : X \to \mathbb{R}$ be continuous on K and let $W \subset X$ be open. Assume that every maximum point of f on K lies in W, i.e.*

$$x \in K \ \text{ and } \ f(y) \le f(x) \ \ \forall y \in K \qquad \Longrightarrow \qquad x \in W.$$

Then there exists $\varepsilon > 0$ with the following property: for every function $g : X \to \mathbb{R}$ such that $|g(x) - f(x)| \le \varepsilon$ for all $x \in K$, every maximum point of g on K lies in W, i.e.

$$x \in K \ \text{ and } \ g(y) \le g(x) \ \ \forall y \in K \qquad \Longrightarrow \qquad x \in W.$$

Proof. Since K is non-empty and compact and f is continuous on K, Weierstrass' theorem (preamble) provides $x_0 \in K$ with $f(y) \le f(x_0)$ for every $y \in K$. Write $M_0 := f(x_0) = \max_K f$.

Case 1: $K \setminus W = \emptyset$. Then $K \subset W$, and $\varepsilon := 1$ works: for any g and any $x \in K$ we have $x \in W$ outright.

Case 2: $K \setminus W \neq \emptyset$. Since W is open, $K \setminus W = K \cap (X \setminus W)$ is the intersection of the compact set K with a closed set, hence compact; and it is non-empty. As f is continuous on K it is continuous on $K \setminus W \subset K$, so Weierstrass' theorem (preamble) provides $x_1 \in K \setminus W$ with

$$f(y) \le f(x_1) \qquad \text{for every } y \in K \setminus W. \tag{1}$$

We claim $f(x_1) < M_0$. Since $x_1 \in K$ we have $f(x_1) \le M_0$. If $f(x_1) = M_0$, then for every $y \in K$ we get $f(y) \le M_0 = f(x_1)$, so x_1 is a maximum point of f on K and the hypothesis forces $x_1 \in W$, contradicting $x_1 \in K \setminus W$. Hence $f(x_1) < M_0$.

Set

$$\varepsilon := \frac{M_0 - f(x_1)}{3} \ > \ 0.$$

Let $g : X \to \mathbb{R}$ satisfy $|g(y) - f(y)| \le \varepsilon$ for all $y \in K$, and let $x \in K$ be such that $g(y) \le g(x)$ for all $y \in K$. Suppose, for contradiction, that $x \notin W$. Then $x \in K \setminus W$, so $f(x) \le f(x_1)$ by (1), and therefore

$$g(x) \;\le\; f(x) + \varepsilon \;\le\; f(x_1) + \varepsilon.$$

On the other hand, taking $y = x_0 \in K$ in the maximality of x for g, and using $|g(x_0) - f(x_0)| \le \varepsilon$,

$$g(x) \;\ge\; g(x_0) \;\ge\; f(x_0) - \varepsilon \;=\; M_0 - \varepsilon.$$

Combining the two displays, $M_0 - \varepsilon \le f(x_1) + \varepsilon$, i.e.

$$M_0 - f(x_1) \;\le\; 2\varepsilon \;=\; \tfrac{2}{3}\big(M_0 - f(x_1)\big),$$

whence $M_0 - f(x_1) \le 0$, contradicting $f(x_1) < M_0$. Therefore $x \in W$, which is the assertion. □

2.17. `HGradContinuousOn`.

Lemma 2.17. *Let $n, m \in \mathbb{N}$, let $\sigma = (\sigma^1, \dots, \sigma^m)$ be smooth vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$, let $\Omega \subset \mathbb{R}^n$ be open and let $u : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on Ω. Then the horizontal gradient $\mathfrak{X}u : \mathbb{R}^n \to \mathbb{R}^m$ is continuous on Ω. This is the step "by assumption and continuity of $\mathfrak{X}u$" at the start of the proof of Proposition `lem:comp0` of the paper.*

Proof. Since u is of class C^2 on the open set Ω, the differential map $y \mapsto Du(y)$, with values in the space $\big(\mathbb{R}^n\big)^*$ of continuous linear functionals on $\mathbb{R}^n$, is continuous on Ω: a C^2 function on an open set is in particular C^1 there, and for a function of class C^1 on an open set the differential is continuous on that set (preamble).

Fix $i \in \{1, \dots, m\}$. By the definition of the horizontal gradient (`HGrad`), the i-th component of $\mathfrak{X}u$ is

$$\big(\mathfrak{X}u(y)\big)_i = Du(y)\big[\sigma^i(y)\big].$$

The map $y \mapsto \sigma^i(y)$ is smooth on $\mathbb{R}^n$, hence continuous, and therefore continuous on Ω. The map $y \mapsto Du(y)$ is continuous on Ω by the previous paragraph. The evaluation pairing

$$\big(\mathbb{R}^n\big)^* \times \mathbb{R}^n \to \mathbb{R}, \qquad (\Lambda, \eta) \mapsto \Lambda[\eta],$$

is continuous, being bilinear and bounded on the finite-dimensional space $\big(\mathbb{R}^n\big)^* \times \mathbb{R}^n$ (preamble). Composing the continuous map $y \mapsto \big(Du(y), \sigma^i(y)\big)$ on Ω with this continuous pairing shows that $y \mapsto Du(y)\big[\sigma^i(y)\big]$ is continuous on Ω.

Thus every component of $\mathfrak{X}u$ is continuous on Ω. A map into the finite-dimensional space $\mathbb{R}^m$ is continuous on a set exactly when each of its m components is (preamble), so $\mathfrak{X}u$ is continuous on Ω. □

2.18. `QuadraticPerturbDeriv`.

Lemma 2.18. *Let $\lambda, u_0 \in \mathbb{R}$ and let*

$$h(s) := s + \lambda\,(s - u_0)^2 \qquad (s \in \mathbb{R}).$$

Then $h \in C^2(\mathbb{R})$ and

$$h'(s) = 1 + 2\lambda\,(s - u_0), \qquad h''(s) = 2\lambda \qquad \text{for every } s \in \mathbb{R}.$$

These are the function h_λ and the two derivative formulas displayed in the proof of Proposition `lem:comp0` of the paper, where it is recorded that $h'_\lambda \ge 1$ on the range of u (because $u - u_0 \ge 0$ there) and that $h''_\lambda = 2\lambda$.

Proof. Write $q(s) := (s - u_0)^2$, so that $h(s) = s + \lambda\, q(s)$.

The derivative of q. The affine map $s \mapsto s - u_0$ is differentiable at every $s \in \mathbb{R}$ with derivative 1, being the sum of the identity (derivative 1) and the constant $-u_0$ (derivative 0) (preamble). Since $q(s) = (s - u_0) \cdot (s - u_0)$, the product rule (preamble) gives, for every $s \in \mathbb{R}$,

$$q'(s) = 1 \cdot (s - u_0) + (s - u_0) \cdot 1 = 2\,(s - u_0).$$

The derivative of h. The identity has derivative 1 at every point, and $s \mapsto \lambda\, q(s)$ has derivative $\lambda\, q'(s) = \lambda \cdot 2(s-u_0)$ by the previous paragraph and the rule for a constant multiple (preamble). Hence, by the sum rule (preamble), h is differentiable at every $s \in \mathbb{R}$ with

$$h'(s) = 1 + \lambda \cdot 2\,(s-u_0) = 1 + 2\lambda\,(s-u_0),$$

which is the first asserted formula.

The second derivative of h. By the previous paragraph the function h' is given by $h'(s) = 1 + 2\lambda\,(s-u_0)$ for every $s \in \mathbb{R}$, i.e. h' is the affine function $s \mapsto 1 + 2\lambda\,(s-u_0)$. Differentiating it at an arbitrary $s \in \mathbb{R}$: the map $s \mapsto s - u_0$ has derivative 1, so $s \mapsto 2\lambda\,(s-u_0)$ has derivative $2\lambda \cdot 1 = 2\lambda$ by the constant-multiple rule, and adding the constant 1 does not change the derivative (preamble). Hence $h''(s) = 2\lambda$ for every $s \in \mathbb{R}$, which is the second asserted formula.

Regularity. The function h is a polynomial in s of degree at most two, namely $h(s) = \lambda s^2 + (1 - 2\lambda u_0)s + \lambda u_0^2$. Polynomials are infinitely differentiable on $\mathbb{R}$ (preamble), so in particular $h \in C^2(\mathbb{R})$. □

3. Main results

3.1. `ComparisonNondeg` (paper, Theorem 1.3).

Theorem 3.1. *Let $n, m \in \mathbb{N}$, let $\sigma = (\sigma^1, \dots, \sigma^m)$ be smooth vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ satisfying the Hörmander rank condition, i.e. such that for every $x \in \mathbb{R}^n$ the linear span of $\{V(x) : V \in \mathrm{HorFamily}(\sigma)\}$ is $\mathbb{R}^n$ (`HormanderRank`), and let $X_i = \sigma^i(x) \cdot \nabla$. Let $\beta \in \mathbb{R}$ and let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $a(\xi) = (a_{i,j}(\xi))_{i,j}$, be continuous on $\mathbb{R}^m \setminus \{0\}$ and such that, for every $\xi \in \mathbb{R}^m \setminus \{0\}$, the matrix $a(\xi)$ is symmetric and positive semi-definite,*

$$\mathcal{E}(\xi) = \big\langle a(\xi)\xi, \xi \big\rangle > 0,$$

and $a(t\xi) = t^\beta a(\xi)$ for every $t > 0$. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $u, v : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on Ω and continuous on $\overline{\Omega}$. Assume that

$$\mathcal{L}u \le 0 \le \mathcal{L}v \quad \text{in } \Omega$$

in the classical sense, i.e. u is a classical sub-solution and v a classical super-solution of $\mathcal{L} = 0$ in Ω (`IsClassicalSubsol`, `IsClassicalSupersol`), and that $u \le v$ on $\partial\Omega$. If

$$\mathfrak{X}u(x) \neq 0 \qquad \text{for every } x \in \operatorname*{argmax}_{\Omega}(u-v) := \Big\{ x \in \Omega :\ (u-v)(x) = \sup_{\Omega}(u-v) \Big\},$$

then $u \le v$ in Ω.

Proof. Suppose, for contradiction, that there is $\bar{x} \in \Omega$ with $u(\bar{x}) > v(\bar{x})$.

Put $K := \overline{\Omega}$ and $f := u - v$. Since Ω is bounded, K is compact (preamble), and $K \neq \emptyset$ because $\bar{x} \in \Omega \subset K$. The function f is continuous on K, since u and v are. By Weierstrass' theorem (preamble) f attains its maximum over K, and

$$M_0 := \max_K f \ \ge\ f(\bar{x}) = u(\bar{x}) - v(\bar{x}) \ >\ 0. \tag{1}$$

Step 1: every maximum point of f on K lies in $U := \{x \in \Omega : \mathfrak{X}u(x) \neq 0\}$. Let $x \in K$ be such that $f(y) \le f(x)$ for all $y \in K$; then $f(x) = M_0 > 0$ by (1). Since $K = \overline{\Omega} = \Omega \cup \partial\Omega$ (preamble), either $x \in \Omega$ or $x \in \partial\Omega$; the latter is impossible because $f = u - v \le 0$ on $\partial\Omega$ by the boundary hypothesis, while $f(x) > 0$. Hence $x \in \Omega$. Moreover $\Omega \subset K$, so $f(y) \le f(x)$ for every $y \in \Omega$, i.e. $(u-v)(x) = \sup_\Omega(u-v)$ and $x \in \operatorname{argmax}_\Omega(u-v)$. The non-degeneracy hypothesis then gives $\mathfrak{X}u(x) \neq 0$, so $x \in U$.

Step 2: U is open. Since u is of class C^2 on the open set Ω, the map $y \mapsto Du(y)$ is continuous on Ω; each σ^i is smooth, hence continuous. By the definition of the horizontal gradient (`HGrad`), its i-th component is $\big(\mathfrak{X}u(y)\big)_i = Du(y)\big[\sigma^i(y)\big]$; this is continuous on Ω, being the composition of the map $y \mapsto \big(Du(y), \sigma^i(y)\big)$, continuous on Ω, with the continuous evaluation pairing. As this holds for every $i \in \{1, \dots, m\}$, the map $\mathfrak{X}u : \mathbb{R}^n \to \mathbb{R}^m$ is continuous on Ω. (This is the paper's step "by assumption and continuity of $\mathfrak{X}u$" at the start of the proof of Proposition `lem:comp0`.)

Now $\mathbb{R}^m \setminus \{0\}$ is open, and

$$U = \Omega \cap (\mathfrak{X}u)^{-1}\big(\mathbb{R}^m \setminus \{0\}\big)$$

is precisely the preimage of $\mathbb{R}^m \setminus \{0\}$ under the restriction of $\mathfrak{X}u$ to Ω. Since that restriction is continuous, U is open in the subspace topology of Ω, i.e. $U = \Omega \cap V$ for some set V open in $\mathbb{R}^n$. Since Ω is itself open in $\mathbb{R}^n$, the intersection $U = \Omega \cap V$ is open in $\mathbb{R}^n$.

Step 3: the perturbed sub-solution. Let $u_0 := \min_K u$ and $u_1 := \max_K u$, which exist by Weierstrass' theorem (preamble). For $\lambda > 0$ set

$$h_\lambda(s) := s + \lambda(s - u_0)^2, \qquad u_\lambda := h_\lambda \circ u.$$

Then h_λ is a polynomial, hence $h_\lambda \in C^2(\mathbb{R})$, with

$$h_\lambda'(s) = 1 + 2\lambda(s - u_0), \qquad h_\lambda''(s) = 2\lambda \quad \text{for all } s \in \mathbb{R}.$$

For $x \in K$ we have $u_0 \le u(x) \le u_1$, hence $u(x) - u_0 \ge 0$ and therefore

$$h_\lambda'\big(u(x)\big) \ge 1, \qquad 0 \;\le\; u_\lambda(x) - u(x) = \lambda\big(u(x) - u_0\big)^2 \;\le\; \lambda\,(u_1 - u_0)^2 \qquad (x \in K). \tag{2}$$

Moreover $u_\lambda = h_\lambda \circ u$ is of class C^2 on Ω (composition of the C^∞ function h_λ with the C^2 function u) and continuous on K.

Step 4: choice of λ and of the perturbed maximum point. The hypotheses of `MaxSetStability` hold with the topological space $\mathbb{R}^n$, the non-empty compact set K, the function $f = u - v$ (continuous on K) and the open set $W := U$ (open by Step 2): by Step 1, every maximum point of f on K lies in U. Hence there is $\varepsilon > 0$ such that

$$\text{for every } g : \mathbb{R}^n \to \mathbb{R} \text{ with } |g - f| \le \varepsilon \text{ on } K, \text{ every maximum point of } g \text{ on } K \text{ lies in } U. \tag{3}$$

Fix $\lambda > 0$ with $\lambda(u_1 - u_0)^2 \le \varepsilon$: if $u_1 = u_0$ take $\lambda = 1$, and otherwise take $\lambda = \varepsilon/(u_1 - u_0)^2 > 0$. Put $g := u_\lambda - v$. By (2), for every $x \in K$,

$$|g(x) - f(x)| = \big|u_\lambda(x) - u(x)\big| \le \lambda(u_1 - u_0)^2 \le \varepsilon.$$

Since g is continuous on the non-empty compact set K, Weierstrass' theorem (preamble) gives $x_\lambda \in K$ with $g(y) \le g(x_\lambda)$ for every $y \in K$. By (3), $x_\lambda \in U$, that is

$$x_\lambda \in \Omega \qquad \text{and} \qquad \mathfrak{X}u(x_\lambda) \neq 0. \tag{4}$$

Step 5: x_λ is a local maximum point of $u_\lambda - v$. The set Ω is open and contains x_λ by (4), and $\Omega \subset K$, so $g(y) \le g(x_\lambda)$ for every $y \in \Omega$. Thus Ω is a neighbourhood of x_λ on which $u_\lambda - v \le (u_\lambda - v)(x_\lambda)$, i.e. x_λ is a local maximum point of $u_\lambda - v$.

Step 6: $\mathcal{L}u_\lambda(x_\lambda) \ge 0$. Set $t := h_\lambda'\big(u(x_\lambda)\big)$; by (2) and $x_\lambda \in K$ we have $t \ge 1 > 0$. Since u is of class C^2 on the open set Ω and $x_\lambda \in \Omega$, u is of class C^2 in a neighbourhood of x_λ; each σ^i is smooth, hence of class C^1 near x_λ; and $h_\lambda \in C^2(\mathbb{R})$. So the first identity of `HGradHHessComp` gives

$$\mathfrak{X}u_\lambda(x_\lambda) = t\,\mathfrak{X}u(x_\lambda) \;\neq\; 0, \tag{5}$$

because $t \neq 0$ and $\mathfrak{X}u(x_\lambda) \neq 0$ by (4).

Now apply `LocalMaxOpIneq` with the open set Ω, the smooth frame σ, the matrix field a (positive semi-definite away from 0 by hypothesis), the functions u_λ and v (both of class C^2 on Ω, by Step 3 and by hypothesis) and the point $x_\lambda \in \Omega$, which is a local maximum point of $u_\lambda - v$ by Step 5 and satisfies $\mathfrak{X}u_\lambda(x_\lambda) \neq 0$ by (5). It gives

$$\mathfrak{X}v(x_\lambda) = \mathfrak{X}u_\lambda(x_\lambda) \neq 0 \qquad \text{and} \qquad \mathcal{L}v(x_\lambda) \le \mathcal{L}u_\lambda(x_\lambda).$$

Since v is a classical super-solution in Ω, $x_\lambda \in \Omega$ and $\mathfrak{X}v(x_\lambda) \neq 0$, the definition of `IsClassicalSupersol` gives $\mathcal{L}v(x_\lambda) \ge 0$. Therefore

$$\mathcal{L}u_\lambda(x_\lambda) \;\ge\; 0. \tag{6}$$

Step 7: $\mathcal{L}u_\lambda(x_\lambda) < 0$. Put $\xi := \mathfrak{X}u(x_\lambda) \neq 0$. Since u is a classical sub-solution in Ω, $x_\lambda \in \Omega$ and $\xi \neq 0$, the definition of `IsClassicalSubsol` gives

$$\mathcal{L}u(x_\lambda) \;\le\; 0. \tag{7}$$

Apply `PerturbationIdentity` with the open set Ω, the smooth frame σ, the matrix field a and the exponent β (the homogeneity hypothesis $a(t\xi) = t^\beta a(\xi)$ for $t > 0$, $\xi \neq 0$ being part of our assumptions), with $h = h_\lambda \in C^2(\mathbb{R})$, $w = u$ of class C^2 on Ω, and $x = x_\lambda \in \Omega$, where $\mathfrak{X}u(x_\lambda) = \xi \neq 0$ and $h'_\lambda\big(u(x_\lambda)\big) = t \geq 1 > 0$. Using $h''_\lambda \equiv 2\lambda$, it gives

$$\mathcal{L}u_\lambda(x_\lambda) = -t^\beta \Big[2\lambda\, \mathcal{E}(\xi) - t\, \mathcal{L}u(x_\lambda)\Big]. \tag{8}$$

We estimate the bracket from below. First, the function $\mathcal{E}$ appearing in the conclusion of `PerturbationIdentity` is the ellipticity function of `EllipticityFn`, namely $\mathcal{E}(\zeta) = \langle a(\zeta)\zeta, \zeta\rangle$, which is exactly the function displayed in the ellipticity hypothesis of the present statement; since $\xi \neq 0$, that hypothesis gives $\mathcal{E}(\xi) > 0$. Together with $\lambda > 0$ this yields $2\lambda\, \mathcal{E}(\xi) > 0$. Second, $t > 0$ and $\mathcal{L}u(x_\lambda) \leq 0$ by (7), so $-t\, \mathcal{L}u(x_\lambda) \geq 0$. Hence

$$2\lambda\, \mathcal{E}(\xi) - t\, \mathcal{L}u(x_\lambda) \;\geq\; 2\lambda\, \mathcal{E}(\xi) \;>\; 0.$$

Since $t > 0$, the real power t^β is positive. Therefore the right-hand side of (8) is the product of the negative number $-t^\beta$ with a positive number, so

$$\mathcal{L}u_\lambda(x_\lambda) \;<\; 0,$$

which contradicts (6).

This contradiction shows that no $\bar{x} \in \Omega$ with $u(\bar{x}) > v(\bar{x})$ exists, i.e. $u \leq v$ in Ω. □

3.2. `CorNocritComparison` (paper, Corollary 1.4, comparison).

Corollary 3.2. *Let $n, m \in \mathbb{N}$, let $\sigma = (\sigma^1, \dots, \sigma^m)$ be smooth vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ satisfying the Hörmander rank condition, i.e. such that for every $x \in \mathbb{R}^n$ the linear span of $\{V(x) : V \in \mathrm{HorFamily}(\sigma)\}$ is $\mathbb{R}^n$ (`HormanderRank`). Let $\beta \in \mathbb{R}$ and let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ be continuous on $\mathbb{R}^m \setminus \{0\}$ and such that, for every $\xi \in \mathbb{R}^m \setminus \{0\}$, the matrix $a(\xi)$ is symmetric and positive semi-definite, $\mathcal{E}(\xi) = \langle a(\xi)\xi, \xi\rangle > 0$, and $a(t\xi) = t^\beta a(\xi)$ for every $t > 0$. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $u, v : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on Ω and continuous on $\overline{\Omega}$, with u a classical sub-solution and v a classical super-solution of $\mathcal{L} = 0$ in Ω (`IsClassicalSubsol`, `IsClassicalSupersol`), and with $u \leq v$ on $\partial\Omega$. If v, or u, has no horizontal critical point in Ω, i.e. if*

$$\mathfrak{X}v(x) \neq 0 \;\text{ for every } x \in \Omega \qquad \text{or} \qquad \mathfrak{X}u(x) \neq 0 \;\text{ for every } x \in \Omega,$$

then $u \leq v$ in Ω.

Proof. All hypotheses of `ComparisonNondeg` except its non-degeneracy hypothesis are among our hypotheses verbatim (the data $\sigma, a, \beta, \Omega, u, v$, the smoothness of σ, the Hörmander rank condition, the continuity, symmetry, positive semi-definiteness, ellipticity and homogeneity of a, the openness and boundedness of Ω, the C^2 regularity of u, v on Ω and their continuity on $\overline{\Omega}$, the sub- and super-solution properties, and the boundary inequality $u \leq v$ on $\partial\Omega$). So it suffices to verify its non-degeneracy hypothesis:

$$(*) \qquad \mathfrak{X}u(x) \neq 0 \qquad \text{for every } x \in \Omega \text{ with } (u-v)(x) = \sup_{\Omega}(u-v).$$

Let $x \in \Omega$ satisfy $(u-v)(y) \leq (u-v)(x)$ for every $y \in \Omega$. We distinguish the two alternatives of the hypothesis.

Case 1: $\mathfrak{X}u(y) \neq 0$ for every $y \in \Omega$. Then in particular $\mathfrak{X}u(x) \neq 0$, since $x \in \Omega$.

Case 2: $\mathfrak{X}v(y) \neq 0$ for every $y \in \Omega$. Since Ω is open and $x \in \Omega$, the set Ω is a neighbourhood of x on which $u - v \leq (u-v)(x)$; hence x is a local maximum point of $u - v$. Moreover u and v are of class C^2 on the open set Ω and $x \in \Omega$, so both are differentiable at x. Therefore `HGradEqAtLocalMax` applies and gives

$$\mathfrak{X}u(x) = \mathfrak{X}v(x),$$

and the right-hand side is non-zero by the hypothesis of this case. Hence $\mathfrak{X}u(x) \neq 0$.

In both cases $\mathfrak{X}u(x) \neq 0$, so $(*)$ holds. Applying `ComparisonNondeg` we conclude that $u \leq v$ in Ω. □

3.3. `CorNocritUniqueness` (paper, Corollary 1.4, uniqueness).

Corollary 3.3. *Let $n, m \in \mathbb{N}$, let $\sigma = (\sigma^1, \dots, \sigma^m)$ be smooth vector fields $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ satisfying the Hörmander rank condition, i.e. such that for every $x \in \mathbb{R}^n$ the linear span of $\{V(x) : V \in \mathrm{HorFamily}(\sigma)\}$ is $\mathbb{R}^n$ (`HormanderRank`). Let $\beta \in \mathbb{R}$ and let $a : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ be continuous on $\mathbb{R}^m \setminus \{0\}$ and such that, for every $\xi \in \mathbb{R}^m \setminus \{0\}$, the matrix $a(\xi)$ is symmetric and positive semi-definite, $\mathcal{E}(\xi) = \langle a(\xi)\xi, \xi\rangle > 0$, and $a(t\xi) = t^\beta a(\xi)$ for every $t > 0$. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $u, v : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on Ω and continuous on $\overline{\Omega}$. Assume that each of u and v is a solution of $\mathcal{L} = 0$ in Ω, i.e. that each of them is simultaneously a classical sub-solution and a classical super-solution in Ω (`IsClassicalSubsol`, `IsClassicalSupersol`); that $u = v$ on $\partial\Omega$; and that v has no horizontal critical point in Ω, i.e. $\mathfrak{X}v(x) \neq 0$ for every $x \in \Omega$. Then*

$$u = v \qquad \text{on } \overline{\Omega}.$$

Proof. All the ambient hypotheses of `CorNocritComparison` on σ, a, β and Ω (smoothness of σ, the Hörmander rank condition, continuity, symmetry, positive semi-definiteness, ellipticity and homogeneity of a, openness and boundedness of Ω) are among our hypotheses verbatim, and both u and v are of class C^2 on Ω and continuous on $\overline{\Omega}$. We apply `CorNocritComparison` twice, exchanging the roles of the two functions.

Step 1: $u \leq v$ in Ω. Apply `CorNocritComparison` with u in the role of the sub-solution and v in the role of the super-solution. Indeed u is a classical sub-solution in Ω and v is a classical super-solution in Ω (both being solutions); the boundary hypothesis holds, since $u = v$ on $\partial\Omega$ gives $u \leq v$ on $\partial\Omega$; and the disjunctive hypothesis holds through its first alternative, namely that the super-solution v has no horizontal critical point in Ω, which is our hypothesis $\mathfrak{X}v(x) \neq 0$ for every $x \in \Omega$. The conclusion is $u(x) \leq v(x)$ for every $x \in \Omega$.

Step 2: $v \leq u$ in Ω. Apply `CorNocritComparison` again, now with v in the role of the sub-solution and u in the role of the super-solution. Indeed v is a classical sub-solution in Ω and u is a classical super-solution in Ω (both being solutions); the boundary hypothesis holds, since $u = v$ on $\partial\Omega$ gives $v \leq u$ on $\partial\Omega$; and the disjunctive hypothesis holds through its second alternative, which for this role assignment reads that the sub-solution v has no horizontal critical point in Ω, again our hypothesis $\mathfrak{X}v(x) \neq 0$ for every $x \in \Omega$. The conclusion is $v(x) \leq u(x)$ for every $x \in \Omega$.

Step 3: conclusion. By Steps 1 and 2, $u(x) = v(x)$ for every $x \in \Omega$. By hypothesis $u(x) = v(x)$ for every $x \in \partial\Omega$. Since $\overline{\Omega} = \Omega \cup \partial\Omega$ (preamble), $u = v$ on $\overline{\Omega}$. □